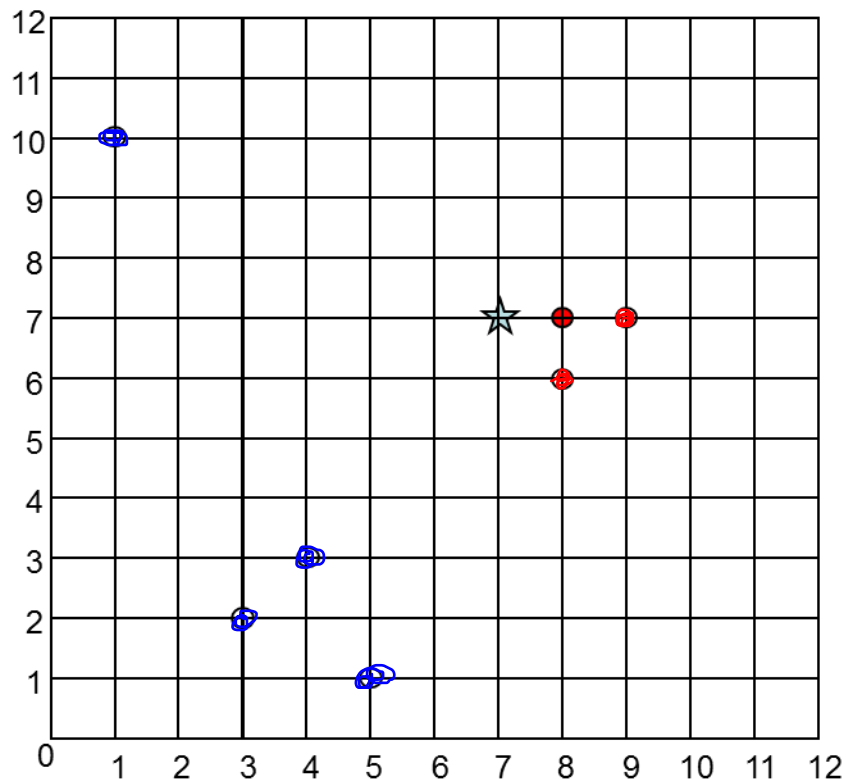


k -means

The following data set with 8 points (2-dimensional feature vectors) is given:



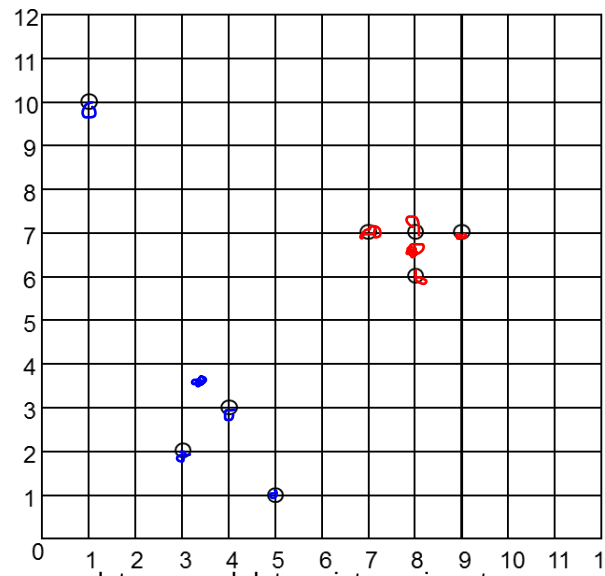
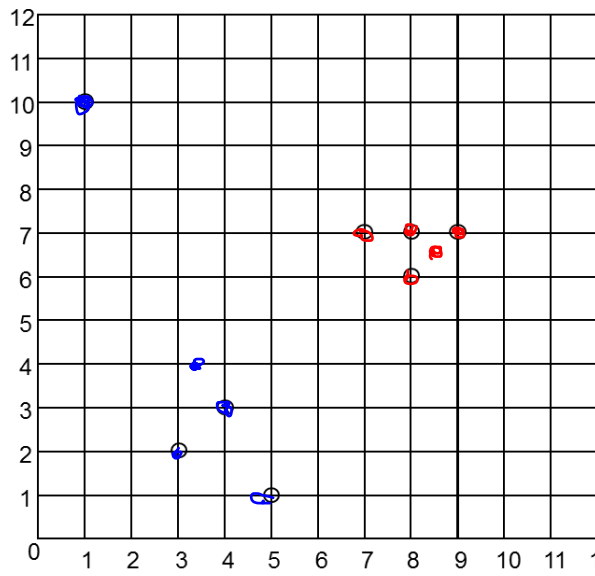
A complete partitioning of the data set into $k = 2$ clusters is to be calculated. The Manhattan distance (L_1 - norm) is used as the distance function between the points.

Create a partition using the k -means method. Use the data points marked by the (blue) asterisk and the filled (red) circle as randomly chosen initial centroids.

Describe each action of the algorithm. After each step, draw the centroids and mark the points according to their cluster affiliation. Remember to use the L_1 -norm when mapping to the centroids.

Hint: Use the templates on the following page (you may not need all templates).

note: single data point in top left pulls centroid in its direction,
so data set should be cleaned before algo usage to prevent this



8/6.75

3.25/4

algo complete: re-eval data points assigns to same centroid

